

AI Knowledge Inbox - Local Setup

Minimal setup and smoke-test instructions for running the interview assignment locally.

1. Prerequisites

Node.js and **Ollama** installed. No external OpenAI API key is required for the current local-model implementation.

```
ollama --version
ollama pull nomic-embed-text
ollama pull gemma3:1b
ollama list
```

2. Clone and configure

```
git clone <GITHUB_REPOSITORY_URL>
cd ai-knowledge-inbox

# Create backend/.env with:
OLLAMA_BASE_URL=http://localhost:11434
OLLAMA_EMBEDDING_MODEL=nomic-embed-text
OLLAMA_LLM_MODEL=gemma3:1b
PORT=3000
DATABASE_PATH=./data/knowledge.db
```

Keep **backend/.env** local and uncommitted. Use **.env.example** as the configuration template.

3. Start the backend

```
cd backend
npm install
npm run build
npm start
```

Expected: **Database initialized and schema applied** and **Server listening on port 3000**.

4. Start the frontend

```
cd frontend
npm install
npm run dev
```

Open **http://localhost:5173** in a browser.

5. Quick smoke test

Test	Expected result
Save a text note	Item appears under Saved Knowledge.
Save a URL	Page is fetched server-side and the item appears.
Ask a relevant question	Answer is returned with a relevant source citation.
Ask an unrelated question	App says it lacks enough saved information and shows no misleading sources.
Restart backend	Previously saved items remain available.

API endpoints: **POST /ingest**, **GET /items**, **POST /query**. The project is intentionally single-user and uses Ollama locally for embeddings and LLM generation.